Olist E-Commerce Data Analysis Report Week

1 – Data Science Internship Assignment

Project Title: Olist E-Commerce Sales and Customer Behaviour Analysis

Objective:
The objective of this project is to analyse the Olist Brazilian E-Commerce dataset to understand sales performance, product category trends, customer purchasing behaviour, delivery performance, and geographic revenue distribution.

The analysis aims to transform raw transactional data into meaningful business insights using data cleaning, exploratory data analysis, key performance indicators (KPIs), and visualizations.

2. Dataset Overview

The Olist Brazilian E-Commerce Public Dataset contains information about online orders, products, customers, sellers, payments, reviews, and deliveries.

The primary tables used in the analysis include:

- **Orders:** Order status, purchase date, and delivery dates.
- **Order Items:** Products purchased, item prices, seller IDs, and freight charges.
- **Customers:** Customer identifiers and geographic information.
- **Products:** Product identifiers and product categories.
- **Category Translation:** Translates product category names into English.
- **Payments:** Payment methods, instalment counts, and payment values.
- **Reviews:** Customer review scores and comments.
- **Sellers:** Seller identifiers and location information.
- **Geolocation:** Geographic coordinates associated with postal-code prefixes.

The tables can be connected using identifiers such as `order_id`, `customer_id`, `product_id`, and `seller_id`. These relationships allow analysis of sales, customer activity, product performance, and delivery outcomes.

```
customers: 99,441 rows | 5 columns
geolocation: 1,000,163 rows | 5 columns
order_items: 112,650 rows | 7 columns
order_payments: 103,886 rows | 5 columns
order_reviews: 99,224 rows | 7 columns
orders: 99,441 rows | 8 columns
products: 32,951 rows | 9 columns
sellers: 3,095 rows | 4 columns
product_category_name_translation: 71 rows | 2 columns
```

3. Cleaning Summary

Data cleaning and preparation were important to improve the reliability of the analysis.

The main activities included:

- Reviewing column names, data types, and table structures.
- Checking for missing values and duplicate records.
- Converting date and timestamp fields into appropriate date formats.
- Examining order statuses and selecting delivered orders for relevant sales KPIs.
- Handling missing delivery timestamps appropriately when calculating delivery performance.
- Reviewing duplicate records in the geolocation table.
- Connecting related tables using common identifiers.
- Defining revenue consistently using product item prices for delivered orders, excluding freight charges.

The previously reviewed geolocation data contained approximately 261,831 duplicate rows out of 1,000,163 records. This duplication should be handled carefully before geographic joins to avoid inflating results.

Missing values were not automatically treated as errors because some fields, such as delivery dates or review comments, may legitimately be unavailable.

```
============================================================
Dataset: customers
============================================================
No missing value found!

============================================================
Dataset: geolocation
============================================================
No missing value found!

============================================================
Dataset: order_items
============================================================
No missing value found!

============================================================
Dataset: order_payments
============================================================
No missing value found!

============================================================
Dataset: order_reviews
============================================================
review_comment_title      87656
review_comment_message    58247
dtype: int64
```

4. KPI Summary

The following KPIs summarize the previously completed analysis.

KPI	Result
Product revenue	Approximately R$13.22 million
Delivered orders	96,478
Average order value	Approximately R$137.04
Units sold	110,197
Average delivery time	Approximately 12.56 days
Late delivery rate	Approximately 8.11%

KPI definitions:

- **Product Revenue:** Sum of item prices for delivered orders, excluding freight.
- **Delivered Orders:** Number of distinct orders marked as delivered.
- **Average Order Value:** Product revenue divided by delivered orders.
- **Units Sold:** Number of delivered order-item records, used as a proxy for units sold.
- **Average Delivery Time:** Average elapsed time between purchase and actual delivery for valid records.
- **Late Delivery Rate:** Percentage of delivered orders with valid delivery dates that arrived after the estimated delivery date.

These figures should be recalculated from the final cleaned data and checked against the Power BI measures before submission.

	KPI	Value
0	Total Revenue	R$ 13,221,498.11
1	Total Orders	96,478
2	Average Order Value	R$ 137.04
3	Total Units Sold	110,197
4	Top-Selling Category	health_beauty
5	Top Revenue-Generating Product	bb50f2e236e5eea0100680137654686c

5. Trend Analysis

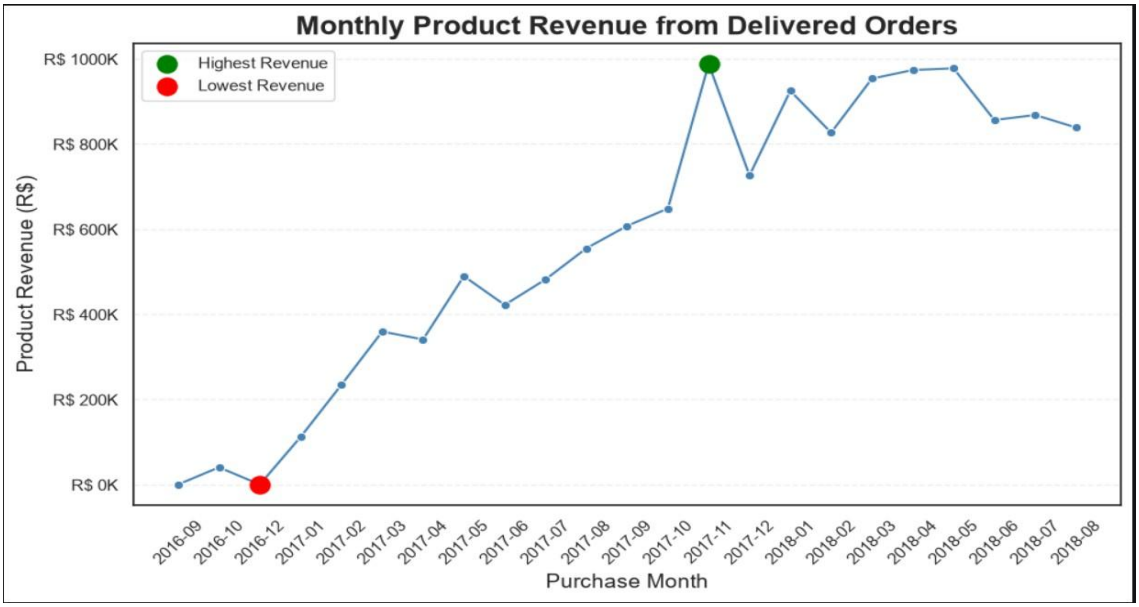
Monthly revenue analysis was used to understand how sales changed over the observed period.

The earlier analysis identified several high-revenue months:

- **November 2017:** Approximately R$987,765.
- **May 2018:** Approximately R$977,545.
- **April 2018:** Approximately R$973,534.

These results indicate periods of relatively strong sales activity. However, the findings should be interpreted carefully because the earliest or latest months in a dataset may contain incomplete observations.

A monthly revenue line chart is suitable for displaying changes over time. Comparing revenue with order volume can also help determine whether sales increases were associated with more orders or higher spending per order.



6. Customer Insights

Customer purchasing behaviour was examined using the unique customer identifier and the number of delivered orders recorded for each customer.

The earlier analysis identified approximately:

- **One-time customers:** 90,557.
- **Repeat customers:** 2,801.

This represents approximately 97% one-time customers and 3% repeat customers among the identified customers in the observed dataset period.

The results suggest that repeat purchasing could be an important area for further investigation. However, these figures represent behaviour within the available data period and should not be interpreted as confirmed lifetime customer retention rates.

Potential strategies include personalized follow-up communications, relevant product recommendations, loyalty incentives, and campaigns aimed at encouraging customers to make a second purchase.

The impact of these strategies should be measured rather than assumed.

7. Visualization Section

Visualizations make the analysis easier to understand and help communicate important business patterns.

The Power BI dashboard should include the following components:

Visualization	Purpose
KPI cards	Display product revenue, delivered orders, average order value, and units sold.
Monthly revenue line chart	Show changes in revenue over time.
Top 10 product categories bar chart	Compare product categories by revenue.
Customer segment donut chart	Compare one-time and repeat customers.
Delivery performance chart	Show on-time/early versus late deliveries.
State revenue bar chart	Identify geographic areas contributing the most revenue.

Dashboard Design

The recommended dashboard layout includes KPI cards at the top, followed by the monthly revenue trend and category revenue ranking. Customer behaviour, delivery performance, and state-level revenue can be displayed in the lower section.

Useful slicers include year, state, and product category, provided that the underlying data model and measures support them correctly.



8. Key Findings

The analysis produced the following important findings:

1. **Sales performance:** Delivered orders generated approximately R$13.22 million in product item-price revenue under the selected definition.
2. **Leading categories:** Health & Beauty generated approximately R$1.23 million, followed by Watches & Gifts at approximately R$1.17 million and Bed, Bath & Table at approximately R$1.02 million.
3. **Geographic concentration:** São Paulo contributed approximately R$5.07 million, or 38.3% of revenue. Rio de Janeiro and Minas Gerais were the next leading states.
4. **Customer behaviour:** Most identified customers placed only one delivered order within the observed period.
5. **Delivery performance:** Average delivery time was approximately 12.56 days, with a late delivery rate of approximately 8.11% among orders with valid delivery and estimated dates.
6. **Data quality:** Duplicate geolocation records and missing timestamps require careful handling to prevent inaccurate calculations.

9. Recommendations

Based on the findings, the following actions are recommended:

- **Improve repeat purchasing:** Test personalized recommendations, post-purchase messages, and loyalty incentives to encourage second purchases.
- **Monitor high-performing categories:** Review inventory availability and marketing performance for leading categories such as Health & Beauty and Watches & Gifts.

- **Investigate delivery delays:** Analyse late deliveries by destination state, seller, and time period to identify potential operational problems.
- **Strengthen geographic analysis:** Maintain visibility into São Paulo while evaluating opportunities to improve sales and service in other states.
- **Monitor KPIs consistently:** Track revenue, order volume, average order value, repeat purchasing, and late delivery rates using clearly defined calculations.
- **Improve data quality:** Document cleaning steps and handle duplicate geolocation records before joining geographic data to other tables.

Profitability should be assessed only if reliable product costs, shipping costs, and other relevant expenses become available.

10. Limitations and Assumptions

The analysis has several limitations:

- The findings are descriptive and do not establish causal relationships.
- Revenue is defined as item price for delivered orders and excludes freight charges.
- Revenue is not equivalent to profit because cost and margin data are not included in the analysis.
- Customer repeat purchasing is measured within the available dataset period, not across a verified lifetime customer history.
- Delivery calculations exclude orders missing required actual or estimated delivery timestamps.
- Some months may contain incomplete data, which can affect comparisons.
- The source dataset represents a historical period and does not necessarily reflect current e-commerce performance.
- The reported figures are working results from the earlier analysis and should be validated against the final cleaned data and dashboard.

Conclusion

The Olist dataset provides useful insights into sales performance, product category contribution, customer purchasing behaviour, delivery reliability, and geographic revenue distribution.

The analysis highlights opportunities to encourage repeat purchasing, investigate delivery delays, and monitor high-performing product categories and regions.

The Power BI dashboard brings these insights together in a visual format that supports easier comparison and interpretation. Clear KPI definitions, careful data cleaning, and transparent documentation of limitations are essential for ensuring that the final report is reliable and professionally presented.